

OBJECT DETECTION FOR LOW VISIBILITY SCENARIOS USING DEEP LEARNING

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CERTIFICATE OF APPROVAL

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DECLARATION

I/we certify that:

- a. The work contained in this project has been done by me under the guidance of my supervisor(s).
- b. The work has not been submitted to any other Institute for any degree or diploma.
- c. I have followed the guidelines provided by the Institute in preparing the project report.
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- e. Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the report.

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LIST OF ABBREVIATIONS

ACDC	Adverse Conditions Dataset with Correspondences
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
BDD100K	Berkeley DeepDrive 100K Dataset
COCO	Common Objects in Context
CSP	Cross Stage Partial Network
CUDA	Compute Unified Device Architecture
DFL	Distribution Focal Loss
DNN	Deep Neural Network
FPS	Frames Per Second
GPU	Graphics Processing Unit
HSV	Hue Saturation Value
IoU	Intersection over Union
mAP	Mean Average Precision
PAN	Path Aggregation Network
RGB	Red Green Blue
SSD	Single Shot Detector
VRAM	Video Random Access Memory
YOLO	You Only Look Once

LIST OF SYMBOLS

%	Percentage
$\times$	Multiplication operator
@	At (used in mAP@50 notation)

ABSTRACT

Object detection plays a central role in computer vision, with direct applications in autonomous driving, traffic monitoring, and surveillance systems. Detection models trained on clear-weather data lose accuracy considerably when tested in fog, rain, night, and snow, where reduced contrast, poor lighting, and environmental distortion alter the visual appearance of objects. Handling this through condition-specific models or extra sensing hardware creates deployment barriers that are difficult to justify at scale.

This work trains a single YOLOv8m detector through a two-step process that moves from general driving knowledge to adverse-condition specificity. BDD100K, containing 100,000 dashcam recordings across mixed weather, first teaches the model to recognize seven road object categories: pedestrian, car, truck, bus, rider, traffic light, and traffic sign. The model weights from this stage then initialize fine-tuning on ACDC, a 4,006-image dataset recorded exclusively in fog, rain, night, and snow. Rather than discarding what was learned, the second stage refines it. Brightness variation, mosaic composition, random erasing, and horizontal flipping run through both stages, keeping the model from depending on visual clarity it will not always have.

Starting from a YOLOv8n baseline that reached mAP@50 of 0.530 and recall of 0.461, the shift to YOLOv8m with sequential training pushed mAP@50 to 0.691 and recall to 0.623, gains of 16.1 and 16.2 percentage points. The car class peaked at mAP@50 of 0.808. At 4.7 ms per image, the system meets real-time requirements without any changes to the network architecture or additions to the sensor setup.

Keywords: Object Detection, YOLOv8, Low Visibility, Adverse Weather, Transfer Learning, BDD100K, ACDC, Autonomous Driving.

PROJECT CO-PO MATRIX

Course Outcomes (COs):

- CO1** Understand the fundamentals of deep learning architectures used for real-time object detection in autonomous driving applications.
- CO2** Apply the YOLOv8 framework for multi-class object detection across diverse weather and lighting conditions using large-scale driving datasets.
- CO3** Design and implement a two-stage transfer learning pipeline incorporating domain-specific fine-tuning for adverse-condition adaptation.
- CO4** Check how well machine learning models work using proper evaluation methods.
- CO5** Develop a robust, real-time capable object detection system.

CO-PO Mapping:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	–	–	–	–	–	–	2
CO2	3	3	3	3	3	–	–	–	–	–	–	2
CO3	3	3	3	3	3	–	–	–	2	–	–	2
CO4	3	2	2	3	2	–	–	–	–	2	–	2
CO5	2	2	3	2	3	2	2	–	2	2	2	3

Scale: 1 – Low, 2 – Medium, 3 – High, – Not Applicable

Program Outcomes (POs):

PO1	Engineering Knowledge
PO2	Problem Analysis
PO3	Design/Development of Solutions
PO4	Conduct Investigations of Complex Problems
PO5	Modern Tool Usage
PO6	The Engineer and Society
PO7	Environment and Sustainability
PO8	Ethics
PO9	Individual and Team Work
PO10	Communication
PO11	Project Management and Finance
PO12	Life-long Learning

CHAPTER 1: INTRODUCTION

1.1 Overview

Object detection is one of the most studied topics in computer vision today. The task involves finding and marking objects in images or video using bounding boxes along with their class labels. Over the last ten years, deep learning models have pushed detection accuracy to very high levels on standard test datasets. These models are now being used in many real world systems including self driving cars, road traffic monitoring, pedestrian safety systems, and security cameras. Architectures like YOLO, Faster RCNN, and SSD have all contributed to this growth. That said, most of these models are built and tested under good lighting and clear weather. When the same models are used in places where visibility is poor due to fog, rain, or darkness, their accuracy falls quite a bit. Closing this gap is both a practical need and an interesting technical challenge.

1.1 Background and Motivation

Detection models that perform well in clear daylight often struggle badly when the scene changes. Fog makes objects harder to see by cutting down contrast and depth. Rain adds streaks and reflections that confuse the model. Nighttime scenes create patchy lighting with very dark shadows that hide important details. Standard training methods do not prepare models for these kinds of inputs. Yet autonomous cars, city surveillance systems, and smart traffic tools all run around the clock in exactly these conditions. Missing a pedestrian on a foggy road or failing to spot a vehicle at night can have very serious consequences. This practical safety concern is what pushed us toward building a detection system that works well across different bad weather situations without needing separate models or special hardware for each one.

1.3 Literature Survey

A number of studies have looked at making object detectors work better in bad weather. One group of researchers added weather aware modules to YOLOv11 and saw gains in foggy and rainy scenes, though their model became quite heavy and was not tested much at night. Another set of experiments combined multiple datasets including ACDC and DAWN with YOLOv8 and found better overall robustness, but did not report much about speed or real time performance. Some work on low visibility scenes used modified YOLO versions but mostly tested on synthetic fog data which does not fully reflect real conditions. A defogging integrated detector showed good results but was too slow for edge devices. Survey papers covering deep learning detectors under rain, fog, and snow

confirmed that all popular models lose significant accuracy in bad weather. Across all reviewed work, there was no lightweight solution that dealt with fog, rain, and nighttime together in a single unified model. This gap in existing research is what the current project aims to fill.

1.4 Problem Statement

Object detection systems are commonly trained and evaluated under normal visibility conditions, where they achieve high accuracy. However, their performance degrades significantly in low visibility environments such as fog, rain, and nighttime scenes due to poor illumination, reduced contrast, and environmental noise. In foggy conditions, distant objects become indistinguishable from the background. Rain introduces visual artifacts like streaks and surface glare that disrupt feature extraction. At night, uneven lighting and deep shadows make it difficult for models to recognize object boundaries and shapes correctly.

Most existing approaches address only a single type of visibility challenge or rely on complex models and external sensors such as LiDAR or infrared cameras, which increases both cost and computational load. These factors limit their effectiveness and real time applicability in standard deployment scenarios. Additionally, models trained on synthetic weather data often fail to generalize well when tested on real world adverse condition footage.

There is therefore a strong need for a unified and practical approach that improves object detection performance consistently across multiple low visibility conditions while maintaining real time speed on standard hardware, without depending on additional sensors or condition specific model architectures.

1.5 Proposed Approach

This project uses a two stage training method built on top of YOLOv8. In the first stage, the model is trained on BDD100K, a large driving dataset with 100,000 images covering both clear and mixed weather scenes. This gives the model a solid base for detecting road objects. In the second stage, the model is further trained on the ACDC dataset, which has real world images taken in fog, rain, snow, and at night. During both stages, augmentation methods like brightness changes, blur, noise, mosaic mixing, and image flipping are used to make the model more adaptable. The whole setup runs on standard GPU hardware without any extra sensors, keeping it practical for real deployment.

1.6 Thesis Objectives

The key objectives of this project are listed below:

1. To study existing research on object detection under low visibility conditions and prepare suitable datasets
2. To implement a baseline object detection model for normal visibility scenarios
3. To evaluate the baseline model under fog, rain, and night conditions
4. To improve detection performance in low visibility environments through training and optimization techniques

1.7 Scope of the Work

This project focuses on object detection for road driving scenarios under adverse visibility conditions. The work targets five commonly encountered road object classes which are car, person, truck, bus, and bike. Two publicly available benchmark datasets are used for training and evaluation, namely BDD100K for general driving scenes and ACDC for real world adverse condition footage. No external sensors, depth maps, or additional hardware inputs are used at any stage, keeping the system accessible and deployable on standard GPU setups.

The project specifically addresses three types of visibility degradation which are fog, rain, and nighttime low light conditions. Other forms of image degradation such as motion blur caused by fast moving vehicles or hardware level sensor noise are outside the scope of this work. The detection pipeline is designed to be lightweight and efficient, and the final model is tested for both detection accuracy and real time inference speed. The goal is to confirm that the proposed approach can run reliably in practical deployment conditions without sacrificing performance across any of the three targeted adverse scenarios.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Overview

This chapter presents a structured review of existing research and patents related to object detection under adverse visibility conditions. The works examined span a wide range of technical approaches, including weather-aware model architectures, large-scale dataset construction, augmentation-based training strategies, multi-sensor fusion systems, and deep learning pipelines designed for real-time deployment. The primary objective of this review is to understand the current state of the field, assess the strengths and limitations of existing methods, and identify the specific research gaps that this project aims to address. The review is organised thematically to allow clearer comparison across related contributions.

2.2 Review of Research Papers

Object detection under poor visibility has received considerable and growing attention in recent years, driven largely by the safety-critical demands of autonomous driving, intelligent traffic surveillance, and urban monitoring systems. When visibility is degraded by fog, rain, snow, or nighttime darkness, standard detectors trained under clear-sky conditions experience significant drops in precision and recall, making robust performance across multiple adverse conditions a practical necessity rather than an academic exercise. The papers reviewed in this chapter are grouped into three broad themes: fog and rain focused detection, nighttime and low-light detection, and multi-condition generalisation.

2.2.1 Fog and Rain Focused Detection

Liu et al. proposed ECL-YOLOv11, an enhanced YOLO variant equipped with dedicated weather-aware modules and evaluated on the ACDC and Foggy Cityscapes datasets [1]. The model demonstrated measurable improvements in detection accuracy under foggy and rainy conditions, benefiting from the additional contextual modules that help the network compensate for contrast loss and texture degradation. However, the architecture introduced considerable model complexity and computational overhead, and the nighttime evaluation subset was not explored in sufficient depth to claim multi-condition generalisability. This work is relevant to the present project as it confirms the effectiveness of weather-aware training on ACDC, though it motivates the pursuit of a lighter and more balanced alternative.

Kumar and Muhammad explored a different strategy, combining YOLOv8 with multi-dataset merging using ACDC and DAWN to improve model robustness across weather

types [2]. By exposing the model to a more diverse training distribution, the authors demonstrated that merged-dataset training yields better generalisation than single-source approaches. However, no detailed frames-per-second or inference latency analysis was reported, limiting the ability to assess the method's real-time suitability for deployment on embedded or vehicle-mounted systems.

Chu et al. introduced D-YOLO, a detection framework that directly integrates a defogging module into the YOLO backbone, trained exclusively on Foggy Cityscapes [4]. While the defogging front-end produced measurable detection gains in visually impaired scenes, the additional computation introduced by the dehazing sub-network rendered the system too expensive for resource-constrained edge devices. The authors themselves acknowledged that a lightweight alternative would be necessary for practical deployment, a gap that the present project attempts to fill through augmentation-based adaptation rather than architectural complexity.

Rajendran et al. conducted a systematic benchmarking study of multiple standard object detectors evaluated on DAWN and Foggy datasets [5]. The study conclusively confirmed that all major models, including Faster R-CNN, SSD, and various YOLO versions, suffer significant and consistent drops in accuracy under adverse weather. While this benchmark is highly valuable as a motivating reference, no improvement methodology was proposed alongside the findings, leaving the benchmarking exercise as a diagnostic contribution rather than a solution.

Majumdar et al. proposed FogGuard, which augmented YOLOv3 with perceptual loss functions to directly optimise for fog-robust feature learning, with evaluation conducted on PASCAL VOC and RTTS [8]. The method improved mAP in foggy conditions but increased training complexity and was constrained by limited diversity in the fog evaluation set, making it difficult to assess how well the approach transfers to real-world dense fog or combined weather conditions. Wang et al. built upon similar ideas with AOYOLO, extending YOLOv8n through defogging augmentation and a WiseIoU loss function, achieving a 3 to 5 percent mAP improvement over the baseline on RTTS and synthetic KITTI benchmarks [10]. Despite these gains, the work remained narrowly focused on fog and relied heavily on synthetically generated data, leaving multi-weather generalisation and real-world ACDC evaluation unexplored.

2.2.2 Nighttime and Low Light Detection

Patil and Kulkarni applied YOLOv8 to a custom low-light road and nighttime dataset, reporting accuracy and speed improvements compared to earlier detection models [11]. While the results were encouraging, the evaluation was limited to an in-house dataset without benchmarking against publicly established low-light benchmarks, and no experiments combining fog with nighttime conditions were conducted. These

limitations leave open the question of how well the approach generalises across different low-light environments.

Jain et al. tested a weather-aware YOLO variant on real road weather data within the context of autonomous vehicle perception and reported performance gains in both foggy and rainy driving scenes [6]. However, the dataset used was relatively limited in its diversity of scene types and geographic coverage, and no standardised evaluation on the ACDC benchmark was carried out, making direct comparison with other methods in the field difficult to establish.

2.2.3 Multi Condition and Survey Based Work

Oussama et al. used a modified YOLOv11 trained on synthetic and real fog and rain data and demonstrated generalized low visibility detection [3]. Real world validation was limited however, and testing on standard datasets like BDD100K was identified as a direction for future work. Mishra and Puhan adapted YOLOv7 for dense fog detection using a modified architecture trained on foggy road footage [7]. Detection accuracy in fog improved but the model remained focused only on fog, and extending it to rain and nighttime using YOLOv8 was noted as a clear gap. Nie et al. conducted a comprehensive survey of deep learning detectors across rain, fog, and snow using multiple referenced datasets [12]. The survey confirmed the adaptability of YOLO based models for adverse weather but noted the use of older YOLO versions and absence of new experiments as limitations. Timofte et al. explored a defog enhanced YOLO variant for road object detection in foggy weather and validated its performance on standard foggy benchmarks [9].

2.3 Review of Patents

Four patents were reviewed that are relevant to the problem of object detection under low visibility and adverse environmental conditions.

The first patent reviewed is US 2023/0298330 A1, filed by Purdue Research Foundation and credited to Chan [13]. It presents a deep learning system for object detection in photon limited environments. The system combines non local feature aggregation with knowledge distillation to extract useful features from images taken under extreme low light conditions, achieving reliable detection at exposure levels as low as one photon per pixel. This work supports the relevance of deep learning driven nighttime detection without additional hardware, which aligns with the camera only approach used in this project.

The second patent, US 11,699,250 B2, proposed by Patel [14], describes a sensor fused perception system that combines camera imagery with radar or LiDAR data to handle fog, rain, snow, and nighttime conditions. When the camera and sensor outputs disagree,

the system generates a virtual object overlay to alert the driver about undetected objects. While this project does not use sensor fusion, this patent confirms the practical importance of robust low visibility detection and establishes the real world challenge that camera only deep learning systems must overcome.

The third patent, US 11,215,999 B2, introduced by Kumar [15], presents an end to end deep learning pipeline designed for multi sensor autonomous driving in degraded conditions including heavy rain, fog, and dust. Its modular preprocessing and unified inference approach reflects a similar philosophy to the two stage training strategy adopted in this project, where BDD100K provides general scene knowledge and ACDC provides adverse condition adaptation.

The fourth patent, US 20180211121 A1, proposed by Smith [16], introduces a nighttime vehicle detection method that converts RGB images to LAB colour space and uses the A channel to extract vehicle contours. A neural network then classifies candidates using headlight and tail light signatures, with synthetic data used during training. This patent confirms that camera only solutions can perform effectively at night through smart feature extraction and deep learning, supporting the design choices made in the present work.

2.4 Comparative Analysis

Table 2.1: Comparative Review of Related Works on Low-Visibility Object Detection

Ref	Core Technology	Relevance and Gap
[1]	Enhanced YOLOv11 with weather aware modules trained on ACDC and Foggy Cityscapes	Improved fog and rain detection but high model complexity and limited nighttime evaluation
[2]	YOLOv8 with multi dataset merging using ACDC and DAWN	Better robustness through merged training but no FPS or real time analysis reported
[3]	Modified YOLOv11 trained on synthetic and real fog and rain data	Generalized low visibility detection but limited real world validation and no BDD100K testing
[4]	Integrated defogging module within YOLO backbone trained on Foggy Cityscapes	Defogging improves detection but computationally expensive and not suitable for edge deployment
[5]	Benchmarking of multiple standard detectors on DAWN and Foggy datasets	Confirms major performance drop in bad weather but proposes no improvement method
[6]	Weather aware YOLO variant tested on real road weather data	Improved fog and rain detection but limited dataset diversity and no standardized ACDC evaluation
[7]	Modified YOLOv7 architecture trained on foggy road dataset	Improved fog detection accuracy but focused only on fog with no extension to rain or nighttime

[8]	YOLOv3 with perceptual loss function trained on PASCAL VOC and RTTS	Improved mAP in fog but limited fog diversity and no lightweight adaptation explored
[9]	YOLOv8n with defogging augmentation and WiseIoU loss on RTTS and synthetic KITTI	3 to 5 percent mAP gain over baseline but focused only on fog using synthetic data
[10]	YOLOv8 trained on custom low light road and nighttime dataset	Better accuracy and speed in low light but no public dataset benchmarking or fog combined scenarios
[11]	Survey of deep learning detectors across rain, fog, and snow	Reviews YOLO adaptability in adverse weather but uses older YOLO versions with no new experiments
[12]	Large scale benchmark dataset for object detection and segmentation	Foundational dataset used for pretraining detection models including YOLO variants
[13]	Non local feature aggregation with knowledge distillation for extreme low light detection	Validates camera only deep learning for nighttime detection but does not address fog or rain
[14]	Camera combined with radar or LiDAR for fog, rain, snow, and nighttime detection	Confirms importance of low visibility detection but depends on additional sensors unlike this project
[15]	CNN based multi sensor pipeline for rain, fog, and dust conditions	Modular pipeline design aligns with two stage training strategy here but relies on multiple sensor inputs
[16]	LAB colour space conversion with neural network classification using headlight and tail light cues	Confirms camera only nighttime detection is viable but limited to night only with no fog or rain coverage

2.5 Research Gaps Identified

After reviewing the existing literature and patents, the following gaps were identified:

1. **Single Condition Focus:** Most studies address only a single weather condition such as fog or rain in isolation, with very few works attempting to handle fog, rain, and nighttime together within one unified model [1][4][7][8].
2. **Sensor Dependency:** Methods that do cover multiple conditions typically depend on complex architectures or additional sensors like LiDAR and radar, making them unsuitable for standard camera based real time deployment [2][6].
3. **Synthetic Data Reliance:** Several studies train and evaluate on synthetic weather data rather than real world adverse condition footage, which reduces confidence in practical generalization [3][10].
4. **Lack of Speed Evaluation:** Real time inference speed is rarely measured or reported alongside detection accuracy, leaving a significant gap in understanding whether these models are actually deployable in live systems [2][11].

5. **Outdated Model Versions:** The use of older YOLO versions such as YOLOv3 and YOLOv7 is still common in recent work, with limited exploration of the improved capabilities offered by YOLOv8 across adverse conditions [7][8][12].
6. **Absence of Progressive Training Strategy:** A two stage training approach using BDD100K for general scene learning followed by domain specific fine tuning on ACDC for adverse condition adaptation has not been widely explored as a standalone lightweight solution.

2.6 Summary

This chapter reviewed existing research and patents covering object detection under challenging visibility conditions including fog, rain, snow, and nighttime environments. The surveyed works confirmed that standard detection models trained under clear conditions lose considerable accuracy when exposed to adverse weather, and that most available solutions are either too condition specific or too hardware dependent to be practical for general real world deployment. Approaches relying on sensor fusion or computationally heavy architectures showed promise in controlled settings but remained difficult to scale for real time camera only applications. Studies using synthetic data for training further highlighted the generalization problem when models are tested on real world footage. Across all reviewed work, a consistent pattern emerged where speed, multi condition robustness, and real world validation were rarely addressed together in a single framework. These collective findings establish a clear motivation for the approach proposed in this project, which combines progressive transfer learning on real world datasets with weather based augmentation to build a unified, lightweight, and practically deployable object detection system for low visibility driving scenarios.

CHAPTER 3: METHODOLOGY

3.1 Overview

This project addresses the performance degradation of object detection systems under adverse visibility conditions specifically fog, rain, and nighttime environments which represent a critical and underexplored challenge in real-world computer vision applications. Standard detection models trained on normal daytime imagery suffer significant accuracy loss when deployed in such conditions due to reduced contrast, poor illumination, motion blur, and environmental noise. This project proposes a unified, camera-only deep learning approach that improves detection robustness across multiple low-visibility conditions simultaneously, without relying on additional sensors or architectural modifications.

The methodology is built around two core strategies: transfer learning for faster convergence and knowledge retention, and weather-aware data augmentation to expose the model to simulated adverse conditions during training. Two publicly available driving datasets were used to support this approach. BDD100K, a large-scale dataset of 100,000 mixed-condition dashcam images, was used to establish a generalizable detection baseline. ACDC, a dedicated adverse-conditions benchmark comprising 4,006 high-resolution images captured exclusively under fog, rain, night, and snow, was subsequently used for targeted domain adaptation.

Two YOLOv8 model variants were evaluated across the training pipeline. YOLOv8n (Nano) served as the initial lightweight baseline, providing a reference performance level against which improvements could be measured. YOLOv8m (Medium) was then adopted as the primary improved model, offering significantly greater representational capacity for learning discriminative features in degraded visual environments. Both models were initialized from COCO pretrained weights and fine-tuned progressively on the project-specific driving datasets using transfer learning.

Training was conducted in two sequential phases. In Phase 1, the model was fine-tuned on BDD100K to develop general-purpose object detection capability across diverse driving conditions including clear, rainy, foggy, and overcast scenes. In Phase 2, the best-performing BDD100K weights served as the initialization point for fine-tuning on ACDC, enabling efficient domain adaptation to low-visibility imagery without discarding the broader detection knowledge acquired in Phase 1. After each training epoch, model performance was evaluated using Precision, Recall, mAP@50, and mAP@50-95, with a validation feedback loop ensuring only the best-performing weights were retained and carried forward. The complete end-to-end pipeline, encompassing data input, preprocessing, model training, iterative evaluation, ACDC fine-tuning, and final output generation, is illustrated in Figure 3.1.

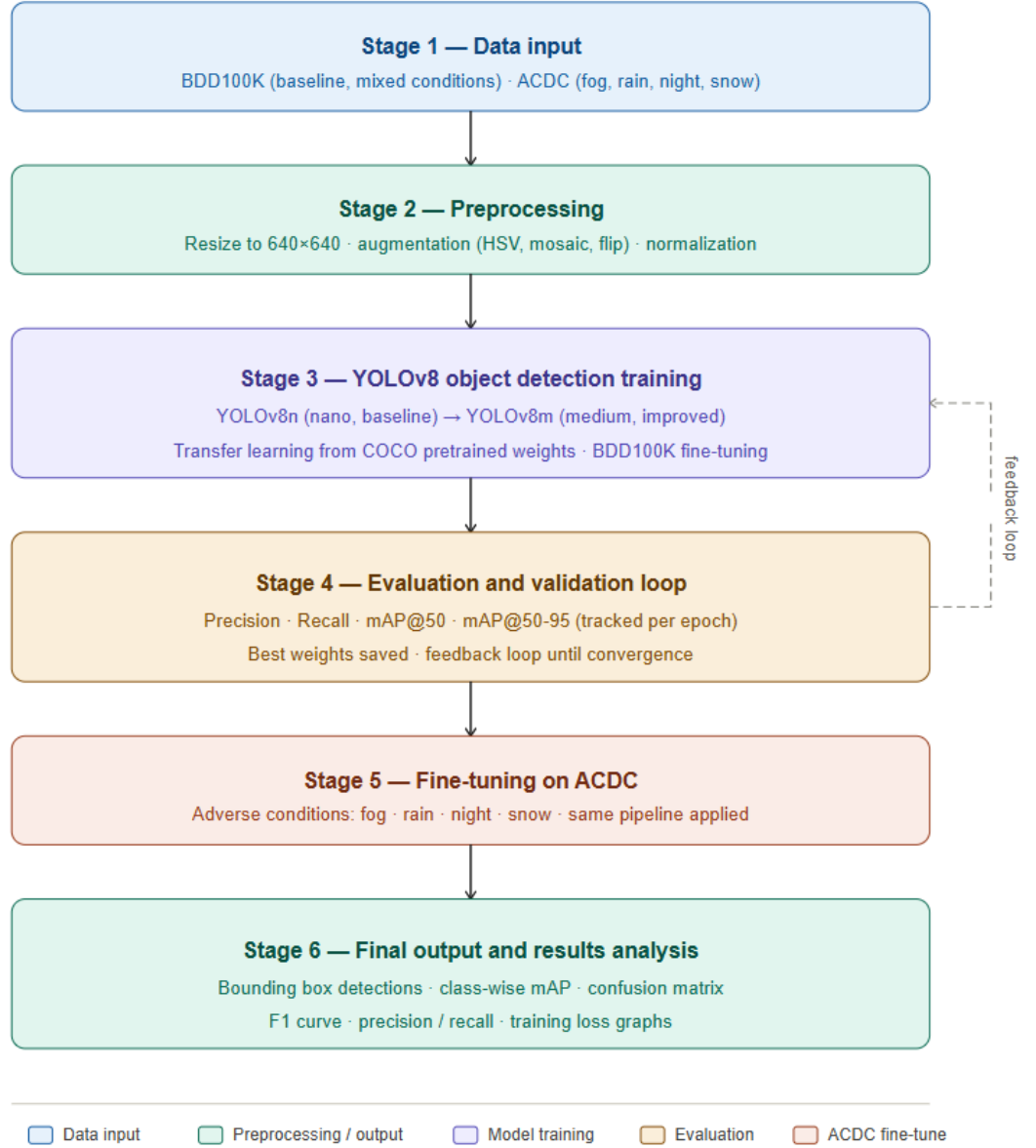


Figure 3.1: End-to-end pipeline for low-visibility object detection using YOLOv8.

3.2 Datasets Used

3.2.1 Dataset 1: BDD100K - Baseline Dataset

The Berkeley DeepDrive 100K (BDD100K) dataset serves as the primary source for baseline model training. It comprises 100,000 high-definition dashcam images at a resolution of 1280×720, capturing real-world urban and highway driving scenes across a diverse range of weather conditions including clear, rainy, foggy, overcast, and snowy environments. For this project, 70,000 images were used for training and 10,000 for validation. The detection classes retained are: Pedestrian, Car, Truck, Bus, Rider, Traffic Light, and Traffic Sign. The scale and weather diversity of BDD100K

make it well-suited to establishing a generalizable detection baseline before adverse-condition fine-tuning.

3.2.2 Dataset 2: ACDC - Adverse Conditions Dataset

The Adverse Conditions Dataset with Correspondences (ACDC) provides 4,006 high-resolution urban street images (1920×1080) captured exclusively under four low-visibility conditions: fog, rain, night, and snow. Approximately 3,000 images were used for fine-tuning and 1,000 for validation. The same object classes as BDD100K were retained to ensure consistency across both training stages. ACDC's exclusive focus on degraded visibility scenarios makes it the primary benchmark for evaluating the core objective of this project.

Table 3.1: Summary of Datasets Used

Parameter	BDD100K	ACDC
Total Images	100,000	4,006
Resolution	1280×720	1920×1080
Weather Conditions	Clear, Rain, Fog, Snow, Overcast	Fog, Rain, Night, Snow
Training Split	70,000	~3,000
Validation Split	10,000	~1,000
Classes Used	Pedestrian, Car, Truck, Bus, Rider, Traffic Light, Traffic Sign	Same as BDD100K

3.3 Experimental Setup

3.3.1 Hardware Configuration

All training and inference experiments were conducted on a local machine with the following hardware specifications:

Table 3.2: Hardware Configuration

Component	Specification
Processor	Intel Core i7 (7th Gen)
GPU	NVIDIA GeForce RTX 5050 Laptop GPU (8 GB VRAM)
RAM	16 GB DDR4
Storage	SSD
System Type	Windows 11 64-bit Laptop

3.3.2 Software Configurations

The software stack used for model development, training, and evaluation is listed below:

Table 3.3: Software Configuration

Software	Version
Operating System	Windows 11
Python	3.10.11
PyTorch	2.12.0 + CUDA
CUDA	13.2
Ultralytics YOLO	YOLOv8 (Ultralytics 8.4.23)
OpenCV	4.x

3.4 Model Architecture

3.4.1 YOLOv8 - You Only Look Once (Version 8)

YOLOv8 (Ultralytics, 2023) was selected as the object detection architecture for this project due to its strong balance between detection accuracy and real-time inference speed. YOLOv8 follows a single-stage detection paradigm, processing an entire image in one forward pass to simultaneously predict bounding boxes and class probabilities making it particularly suitable for latency-sensitive applications such as autonomous driving perception.

The YOLOv8 architecture consists of three principal components: a CSPDarknet-based backbone for hierarchical feature extraction, a Path Aggregation Network (PAN) neck for multi-scale feature fusion, and a decoupled detection head that independently predicts objectness, class probabilities, and bounding box coordinates. This architectural design enables the model to detect objects across a wide range of scales simultaneously, which is especially important in driving scenarios where pedestrians and vehicles appear at vastly different distances and sizes.

Two model variants were evaluated during experimentation:

1. **YOLOv8n (Nano):** A lightweight model trained over 50 epochs on BDD100K. This served as the initial baseline configuration.
2. **YOLOv8m (Medium):** A larger model subsequently fine-tuned on BDD100K and then on ACDC. This forms the final improved model.

3.4.2 Training Pipeline

The training pipeline consisted of two sequential phases:

Phase 1 - Baseline Training (BDD100K): The YOLOv8 model was fine-tuned on BDD100K to learn general object detection features across mixed driving conditions. After convergence, the best model weights were saved for subsequent evaluation and fine-tuning.

Phase 2 - Low-Visibility Fine-Tuning (ACDC): The trained BDD100K weights were used as the starting point for fine-tuning on the ACDC dataset, which contains exclusively fog, rain, night, and snow imagery. This two-stage approach ensures that the model retains general detection knowledge while adapting specifically to low-visibility feature distributions.

3.5 Preprocessing and Data Augmentation

3.5.1 Image Preprocessing

All input images were resized to a standardized resolution of 640×640 pixels prior to training, regardless of their native resolution (1280×720 for BDD100K; 1920×1080 for ACDC). This standardization ensures consistent tensor dimensions across batches and is compatible with YOLOv8's anchor-based detection grid. Pixel values were normalized to the $[0, 1]$ range to ensure numerical stability during gradient computation.

3.5.2 Data Augmentation Strategy

A comprehensive set of augmentation techniques was applied during training to simulate adverse driving conditions and improve model generalization. These augmentations are applied stochastically at each training iteration, ensuring the model is never exposed to identical transformed versions of the same image across epochs.

3.5.3 Image Preprocessing

All input images were resized to a standardized resolution of 640×640 pixels prior to training, regardless of their native resolution (1280×720 for BDD100K; 1920×1080 for ACDC). This standardization ensures consistent tensor dimensions across batches and is compatible with YOLOv8's anchor-based detection grid. Pixel values were normalized to the $[0, 1]$ range to ensure numerical stability during gradient computation.

Table 3.4: Data Augmentation Techniques and Parameters

Augmentation Technique	Parameter/Value	Purpose
HSV Value (Brightness)	hsv_v = 0.4	Simulates varying illumination and night conditions
HSV Saturation	hsv_s = 0.7	Colour variation under different weather
Horizontal Flip	fliplr = 0.5	Spatial diversity across road directions
Scaling / Zoom	scale = 0.5	Simulates objects at varying distances
Mosaic Augmentation	mosaic = 1.0	Multi scale context and small object detection
Random Erasing	erasing = 0.4	Occlusion robustness
Auto Augmentation	RandAugment	General purpose stochastic augmentation
Translation	translate = 0.1	Camera perspective variation

3.6 Training Configuration

3.6.1 Phase 1 - BDD100K Training (YOLOv8n / YOLOv8m)

Table 3.5: Training Hyperparameters - Phase 1 (BDD100K)

Parameter	Value
Epochs	50
Image Size	640 × 640
Optimizer	Auto (AdamW)
Initial Learning Rate (lr0)	0.01
Final Learning Rate (lrf)	0.01
Momentum	0.937
Weight Decay	0.0005
Warmup Epochs	3
IoU Threshold	0.7
Batch Size	Auto
AMP (Mixed Precision)	Enabled
Pretrained Weights	Yes (COCO)

3.6.2 Phase 2 - ACDC Fine-Tuning (YOLOv8m)

Table 3.6: Training Hyperparameters - Phase 2 (ACDC Fine-Tuning)

Parameter	Value
Epochs	20
Image Size	640 × 640
Batch Size	3
Device	GPU (CUDA:0)
Workers	1
Pretrained Weights	BDD100K trained weights
IoU Threshold	0.7
AMP (Mixed Precision)	Enabled

The reduced batch size of 3 during ACDC fine-tuning was a practical constraint imposed by the higher native resolution of ACDC images (1920×1080) and the 8 GB VRAM limit of the RTX 5050. A single worker was used to minimize CPU-GPU data transfer overhead given these memory constraints.

3.7 Evaluation Metrics

Model performance was assessed using the following standard object detection metrics, computed per class and aggregated across all classes on the respective validation sets:

1. **Precision (P):** The proportion of predicted bounding boxes that correctly correspond to a ground truth object. A higher value indicates fewer false positives.
2. **Recall (R):** The proportion of ground truth objects successfully detected by the model. A higher value indicates fewer missed detections.
3. **mAP@50:** Mean Average Precision computed at an IoU threshold of 0.50. This is the primary benchmark metric for object detection performance comparison.
4. **mAP@50-95:** Mean Average Precision averaged across IoU thresholds from 0.50 to 0.95 in steps of 0.05, providing a stricter measure of bounding box localization accuracy.
5. **F1 Score:** The harmonic mean of Precision and Recall, used to assess the overall balance between detection completeness and correctness.

The F1-Confidence curve was additionally analyzed to identify the optimal operating confidence threshold for each model configuration, which is particularly relevant for safety-critical deployment where the trade-off between false positives and missed detections must be carefully calibrated.

CHAPTER 4: WORK DONE

4.1 Overview of Work Completed

The project was executed in four progressive phases as outlined in the project plan. As of the current reporting period, Phase 1 through Phase 3 have been fully completed, covering literature review, dataset preparation, environment setup, baseline model training, and low-visibility model improvement through domain-specific fine-tuning. The following subsections document the work completed at each phase in detail.

4.2 Phase 1: Dataset Collection and Environment Setup

The first phase involved preparing the computational environment and organizing both datasets for training. The following tasks were completed:

1. Python 3.10.11 environment configured with PyTorch 2.12.0, CUDA 13.2, and Ultralytics YOLOv8 (v8.4.23). GPU driver compatibility was verified and CUDA acceleration confirmed operational on the NVIDIA RTX 5050 Laptop GPU.
2. BDD100K dataset downloaded and converted from native JSON format to YOLO annotation format. Annotation labels were mapped to the seven target classes: Pedestrian, Car, Truck, Bus, Rider, Traffic Light, and Traffic Sign.
3. ACDC dataset organized by weather condition (fog, rain, night, snow) and converted to YOLO-compatible label format, maintaining class mapping consistency with BDD100K.
4. Configuration YAML files prepared for both datasets specifying class mappings, training and validation directory paths, and the number of detection targets.
5. Initial exploratory analysis of class distributions in BDD100K revealed significant instance imbalance, with Pedestrian instances (91,349) substantially outnumbering Bus (7,210) and Truck (11,672) instances an observation that informed subsequent augmentation strategy decisions.

4.3 Phase 2: Baseline Model Training on BDD100K (YOLOv8n)

The initial baseline model was trained using the YOLOv8n (Nano) architecture on the BDD100K dataset for 50 epochs. This phase established the reference performance level against which all subsequent improvements are measured.

Key observations from this training run:

- Training completed in approximately 19 hours on the RTX 5050 GPU, with all three loss components (box, classification, and DFL) showing consistent downward trends, confirming stable convergence throughout.
- The baseline model achieved a validation mAP@50 of **0.530** across all classes, with the Car class achieving the strongest individual performance at mAP@50 of 0.582.
- Recall was limited at 0.461, reflecting difficulty detecting objects in darker and

degraded image regions where foreground-background contrast is reduced.

- Inference speed was recorded at 2.5 ms per image, confirming real-time deployment viability even at the baseline configuration.

The dataset distribution analysis revealed significant class imbalance within BDD100K. The Pedestrian class dominated with 91,349 instances, substantially outnumbering Car (29,971), Truck (11,672), and Bus (7,210). This imbalance directly influenced per-class detection performance, with the Bus class consistently exhibiting the weakest recall and mAP scores across the baseline evaluation.

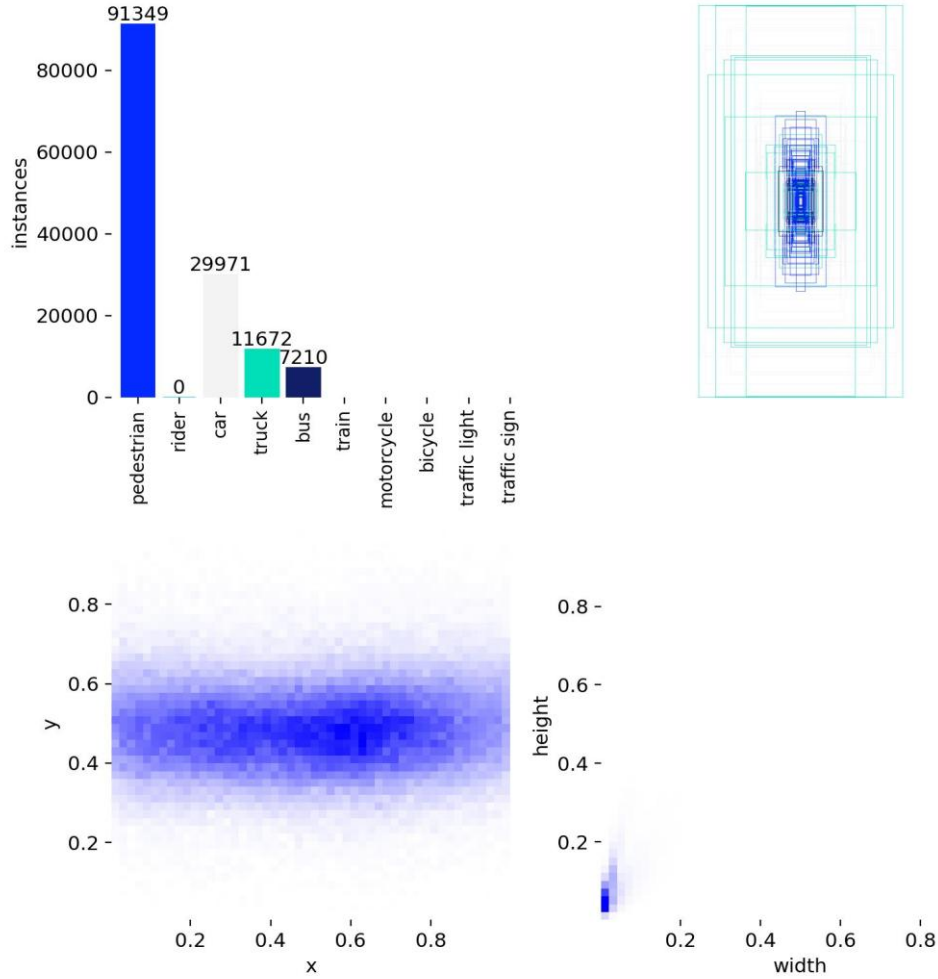


Figure 4.1: BDD100K class distribution and bounding box statistics

4.4 Phase 3: Model Improvement - YOLOv8m on BDD100K and ACDC

To address the limitations of the small model baseline, the architecture was upgraded to YOLOv8m (Medium), which provides significantly greater parameter capacity for learning complex low-visibility features. This phase consisted of two sequential sub-stages.

4.4.1 Stage 1 : YOLOv8m Training on BDD100K

The YOLOv8m model was trained on BDD100K for 50 epochs with the class set expanded to include Traffic Light and Traffic Sign, bringing the total to seven detection categories. The augmentation strategy described in Section 3.5 was applied throughout. Notable improvements over the YOLOv8s baseline:

- Overall mAP@50 improved from 0.530 to **0.691**, a gain of 16.1 percentage points.
- Precision improved from 0.658 to 0.740, and Recall improved from 0.461 to 0.623.
- The Car class achieved the strongest mAP@50 of 0.808, followed by Traffic Sign (0.702) and Traffic Light (0.666), demonstrating the medium model's improved capacity for detecting smaller and more visually complex object categories.

4.4.2 Stage 2 : Fine-Tuning YOLOv8m on ACDC

Fine-tuning ran for 20 epochs using BDD100K-trained weights as initialization, with batch size 3 to accommodate GPU memory constraints.

Key observations:

- Loss curves exhibited a characteristic initial transient increase at epochs 1–3, reflecting domain shift from BDD100K to ACDC, before resuming a consistent downward trend.
- Validation mAP@50 rose progressively across all 20 epochs, confirming successful domain adaptation.
- Precision stabilized around 0.740 and Recall showed a steady upward trend in later epochs.

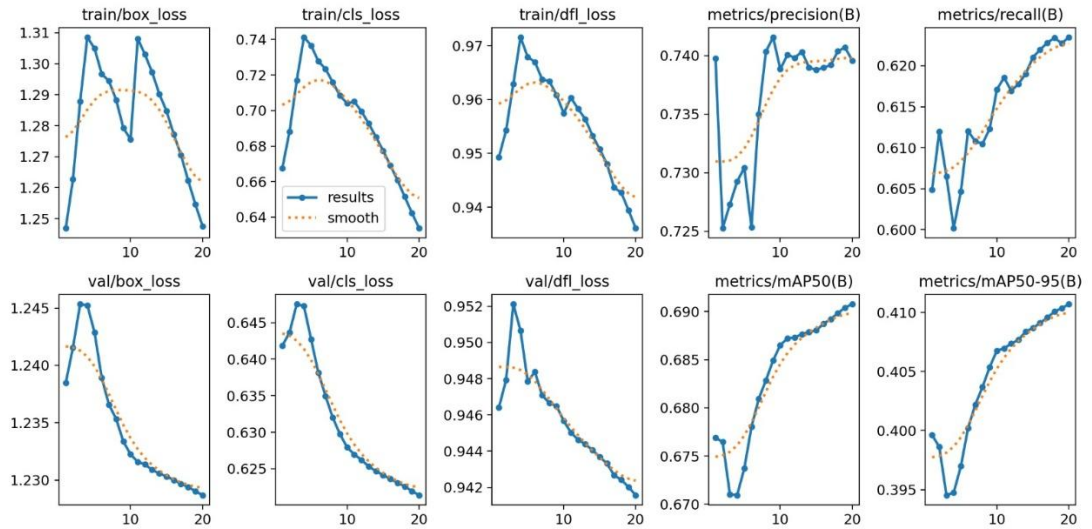


Figure 4.2: YOLOv8m ACDC fine-tuning loss and metric curves

4.4.3 Sample Detection Outputs - BDD100K Validation

Detection results confirm that the model successfully identifies Pedestrians, Cars, Trucks, and Buses across a range of lighting and weather conditions. Some missed detections and overlapping label instances are observed in dense nighttime scenes, which is expected at this stage and is addressed through the ACDC fine-tuning stage.

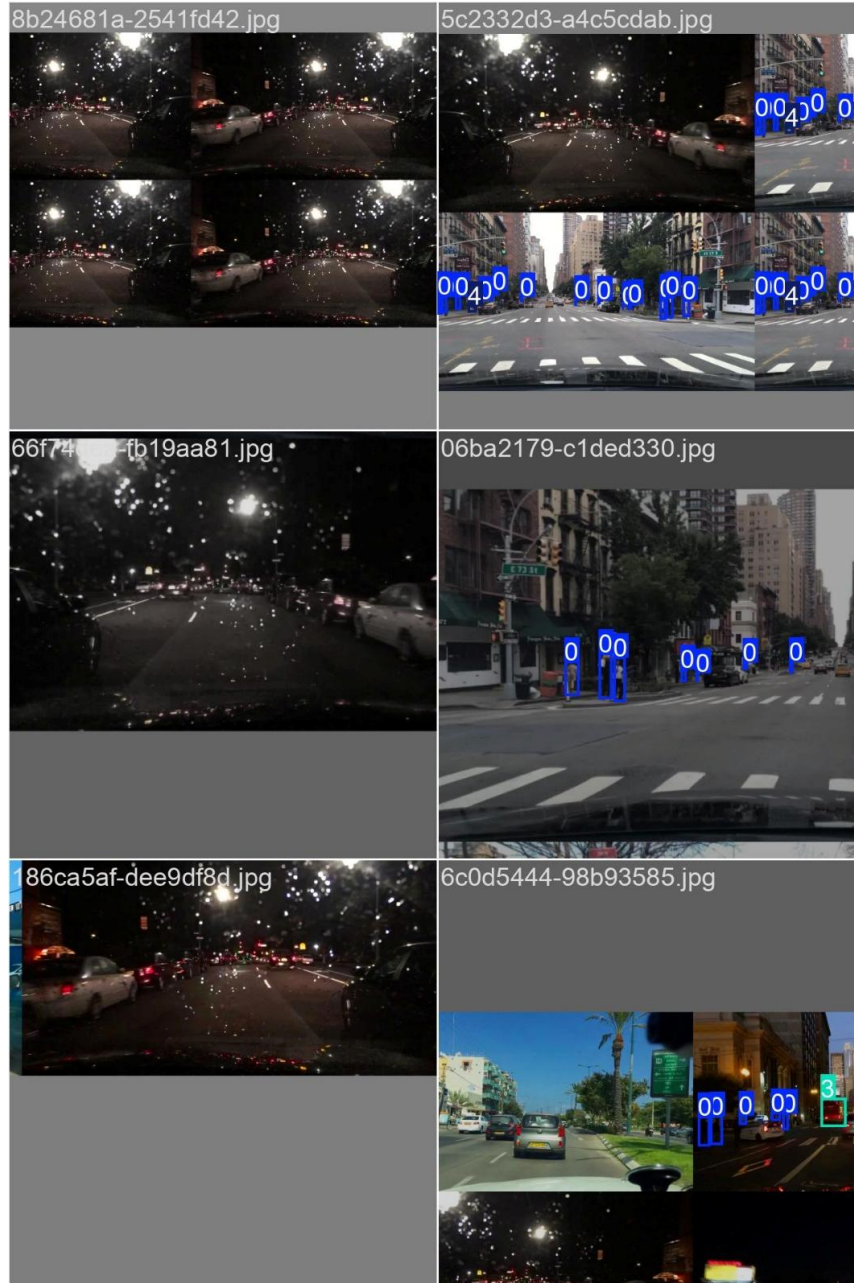


Figure 4.3: BDD100K ground truth annotations

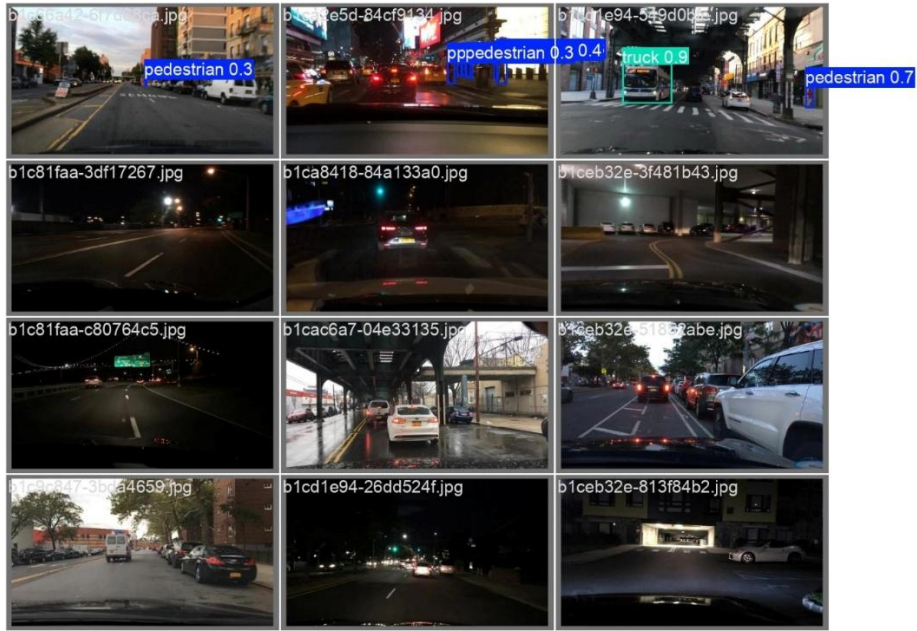


Figure 4.4: YOLOv8m predictions on BDD100K validation images

4.4.4 Sample Detection Outputs - ACDC Validation

The model demonstrates the ability to locate and classify objects across severely degraded nighttime and rainy scenes, producing bounding boxes with reasonable confidence scores even where ambient illumination is near-absent or precipitation causes significant windshield distortion.

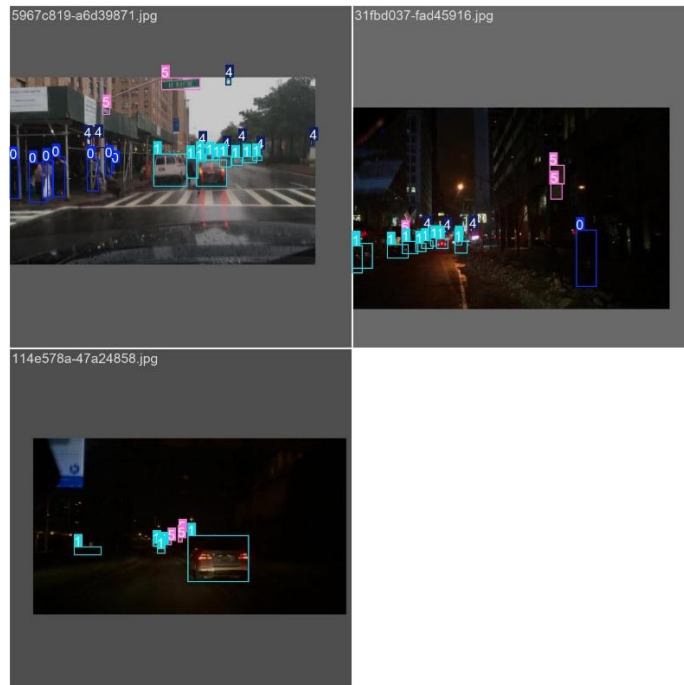


Figure 4.5: YOLOv8m detection on ACDC adverse condition scenes

CHAPTER 5: RESULTS

5.1 Overview

This chapter presents the quantitative and qualitative evaluation results from both training phases. Results are organized as: (1) YOLOv8n Baseline on BDD100K, (2) YOLOv8m Improved Model on BDD100K, and (3) YOLOv8m Fine-Tuned on ACDC. Evaluation metrics include Precision, Recall, mAP@50, mAP@50-95, F1 Score, confusion matrices, and training curve analysis. All metrics were computed on the respective held-out validation splits and reflect the best model weights saved during training based on validation mAP@50.

5.2 YOLOv8n Baseline Results (BDD100K - 50 Epochs)

5.2.1 Quantitative Performance

The YOLOv8n baseline model was evaluated on the 10,000-image BDD100K validation split following 50 epochs of training.

Table 5.1: YOLOv8s Baseline Validation Results (BDD100K)

Class	Images	Instances	Precision	Recall	mAP@50
Pedestrian	3,220	13,262	0.716	0.492	0.578
Car	2,689	4,245	0.682	0.500	0.582
Truck	1,242	1,597	0.676	0.480	0.564
Bus	578	1,007	0.558	0.374	0.396
All Classes	10,000	20,111	0.658	0.461	0.530

Inference speed: **2.5 ms per image**, confirming real-time deployment suitability.

5.2.2 Training Curve

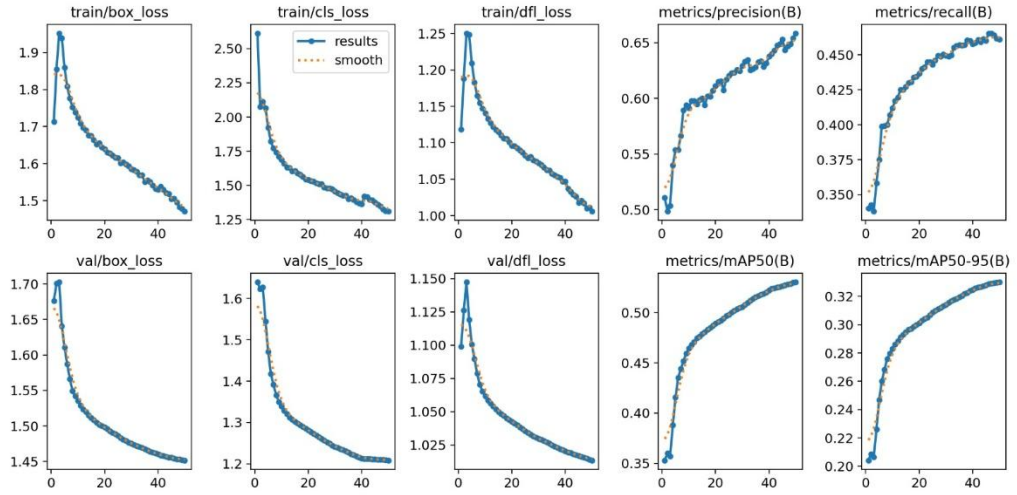


Figure 5.1: YOLOv8n training and validation curves (50 epochs)

Loss curves demonstrate consistent convergence across all three components. Precision increased steadily from ~ 0.50 at epoch 1 to 0.658 at epoch 50. The absence of divergence between training and validation curves confirms no overfitting. The continued upward metric trend at epoch 50 indicates that extended training would likely yield further gains.

5.2.3 F1-Confidence Curve

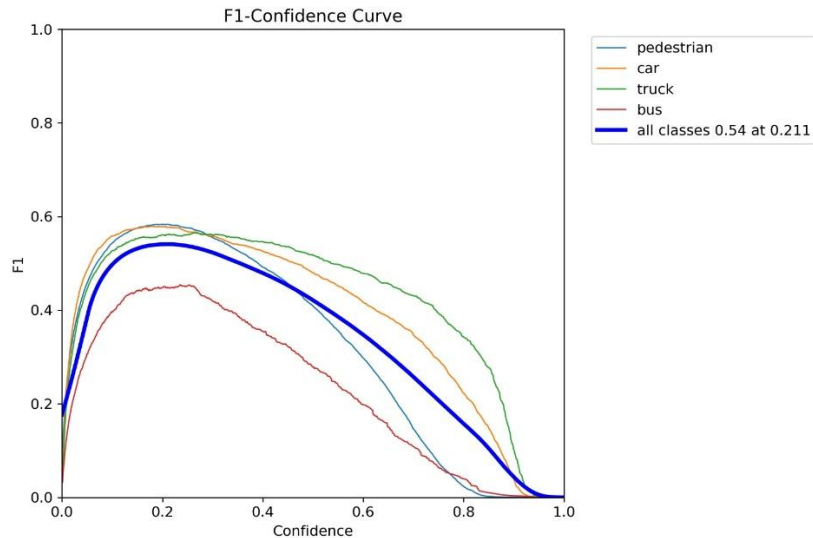


Figure 5.2: YOLOv8n F1-Confidence curve

The F1 curve peaks at **0.54** at a confidence threshold of 0.211. The low optimal threshold reflects the model's tendency to assign moderate confidence scores in degraded image regions, particularly for pedestrians in nighttime scenes.

5.2.4 Precision-Recall Curve

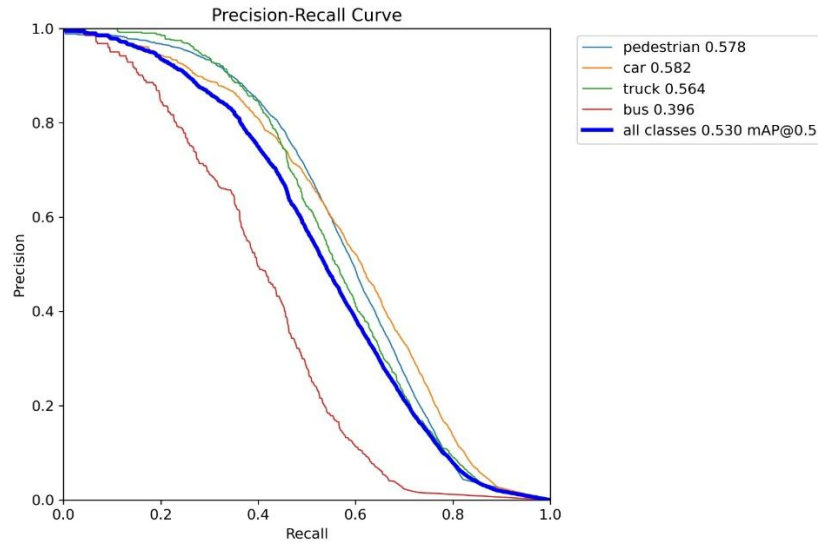


Figure 5.3: YOLOv8n Precision-Recall curve

The Truck class retains high precision across moderate recall values, while Bus exhibits the steepest precision decline directly attributable to its severe underrepresentation in training data (7,210 vs. 91,349 Pedestrian instances).

5.2.5 Confusion Matrix

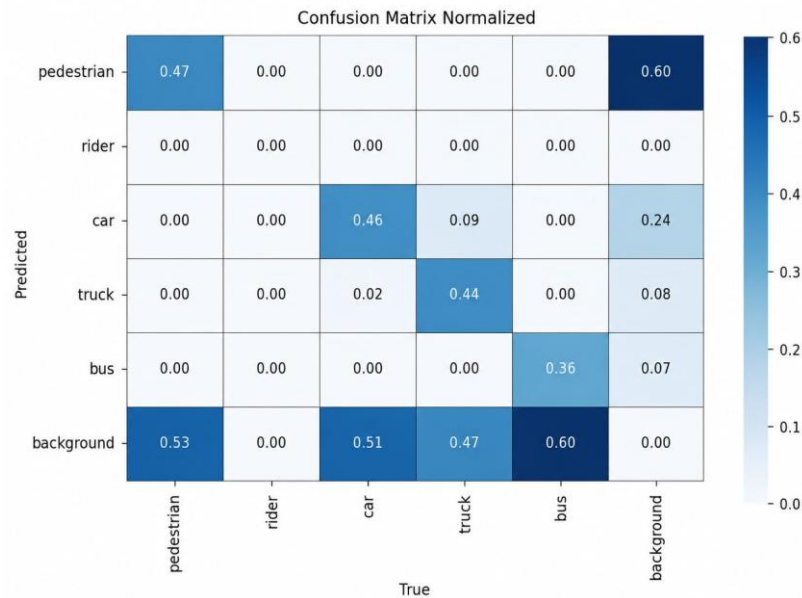


Figure 5.4: YOLOv8n normalized confusion matrix

The normalized confusion matrix reveals key patterns. Pedestrians are correctly detected at 0.47, with 0.53 misclassified as background. Car and Truck show true positive rates of 0.46 and 0.44 respectively, with notable inter-class confusion (Car-Truck: 0.09). Bus records the lowest true positive rate of 0.36, consistent with its mAP@50 of 0.396.

5.3 YOLOv8m Improved Model Results (BDD100K)

5.3.1 Quantitative Performance

Table 5.2: YOLOv8m Validation Results - BDD100K (7 Classes)

Class	Images	Instances	Precision	Recall	mAP@50	mAP@50-95
Person	3,220	13,262	0.781	0.552	0.650	0.333
Car	9,879	102,506	0.811	0.741	0.808	0.507
Truck	2,689	4,245	0.692	0.594	0.668	0.487
Bus	1,242	1,597	0.718	0.562	0.650	0.507
Traffic Light	5,653	26,885	0.705	0.647	0.666	0.255
Traffic Sign	8,221	34,908	0.731	0.645	0.702	0.375
All Classes	10,000	183,403	0.740	0.623	0.691	0.411

5.3.2 Comparison: YOLOv8s vs. YOLOv8m

Table 5.3: Comparison - YOLOv8s Baseline vs. YOLOv8m Improved Model

Metric	YOLOv8s (Baseline)	YOLOv8m (Improved)	Change
Precision	0.658	0.740	+0.082
Recall	0.461	0.623	+0.162
mAP@50	0.530	0.691	+0.161
mAP@50-95	0.330	0.411	+0.081
F1 Score	0.54	0.67	+0.013

The upgrade from YOLOv8s to YOLOv8m yielded a 16.1 percentage point gain in mAP@50 and a 16.2 percentage point improvement in Recall, validating the benefit of increased model capacity for learning complex low-visibility feature representations.

5.3.3 Training Curves

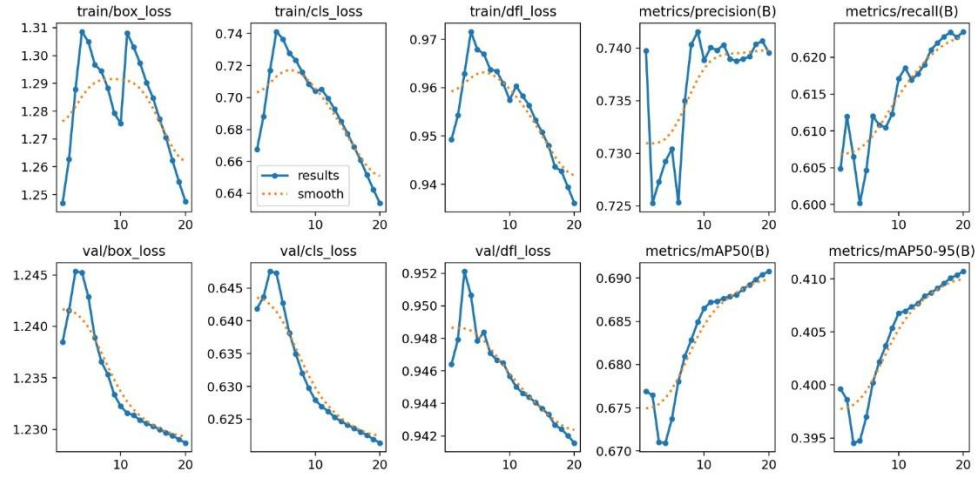


Figure 5.5: YOLOv8m training and validation curves (BDD100K)

5.3.4 F1-Confidence Curve

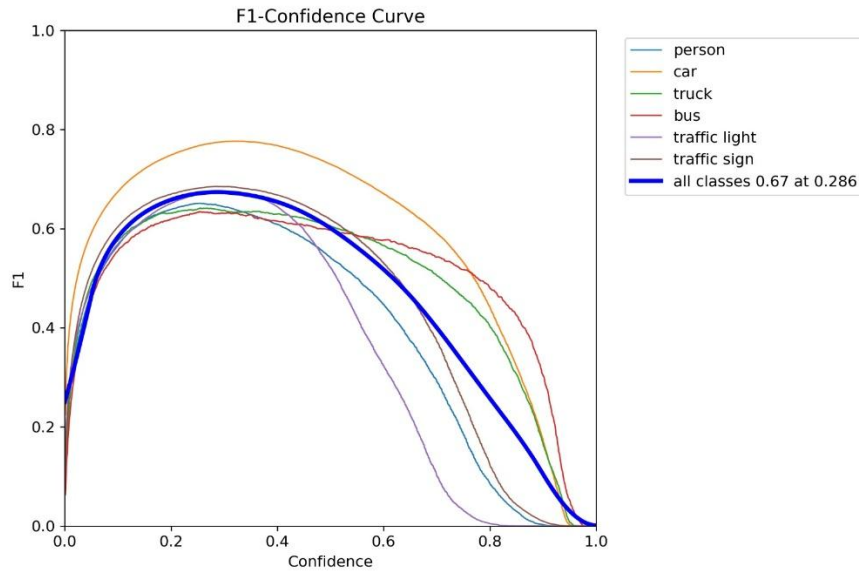


Figure 5.6: YOLOv8m F1-Confidence curve

The peak F1 rises to **0.67** at confidence 0.286, compared to 0.54 at 0.211 for the baseline, reflecting improved confidence calibration and stronger feature activations from the medium model.

5.3.5 Confusion Matrix

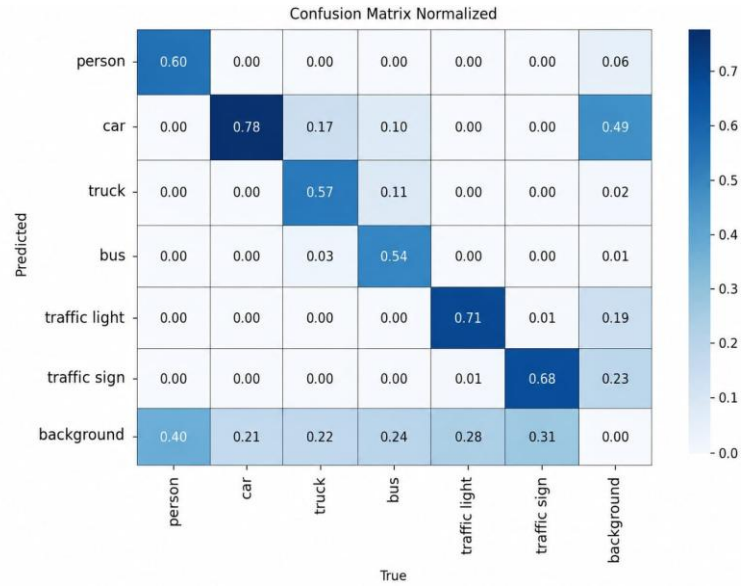


Figure 5.7: YOLOv8m normalized confusion matrix

The YOLOv8m confusion matrix shows clear improvement across all classes. Car achieves 0.78, Traffic Light 0.71, Traffic Sign 0.68, and Person 0.60 (up from 0.47 in baseline). Background misclassification for Person (0.40) persists, attributable to pedestrian silhouette ambiguity under low illumination. Truck-Car (0.17) and Bus-Truck (0.11) inter-class confusion remain the dominant misclassification patterns.

5.3.6 Dataset Distribution

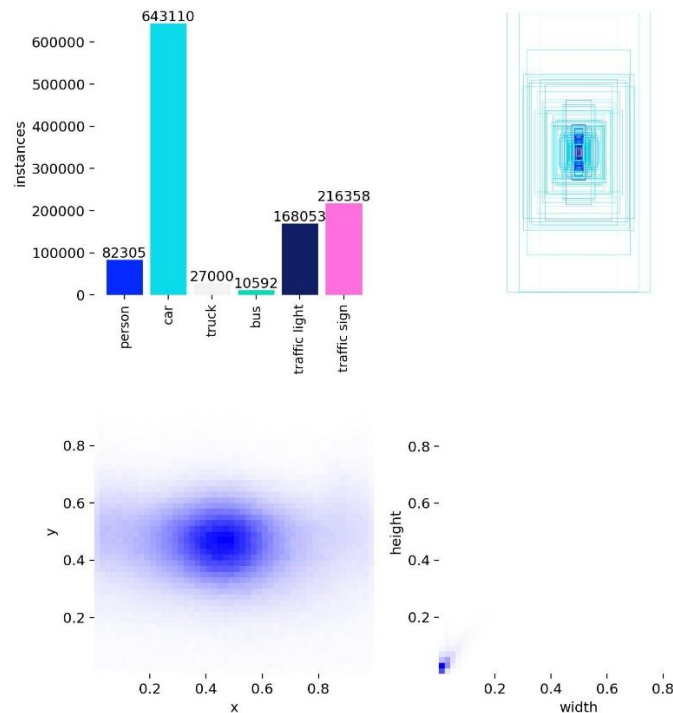


Figure 5.8: YOLOv8m BDD100K dataset distribution

Car dominates with 643,110 instances, followed by Traffic Sign (216,358) and Traffic Light (168,053), directly explaining the consistently higher detection scores for these classes across all evaluation metrics.

5.4 YOLOv8m Fine-Tuned Results (ACDC — Low-Visibility)

5.4.1 Training Progression on ACDC

Following 20 epochs of fine-tuning on ACDC, the model exhibited stable convergence with the following key characteristics:

- Box, Classification, and DFL losses trended downward across all 20 epochs, with a characteristic transient spike at epochs 1–3 reflecting the distributional shift from BDD100K's mixed-condition imagery to ACDC's exclusively degraded-visibility scenes.
- Validation mAP@50 rose progressively from approximately 0.674 at epoch 1 to **0.691** at epoch 20, confirming continued improvement in adverse-condition detection capability throughout the fine-tuning run.
- Precision stabilized around 0.740 in later epochs, while Recall showed a steady upward trend to 0.623, indicating the model's growing ability to locate objects in visually degraded scenes.

5.4.2 ACDC Validation Detection Outputs

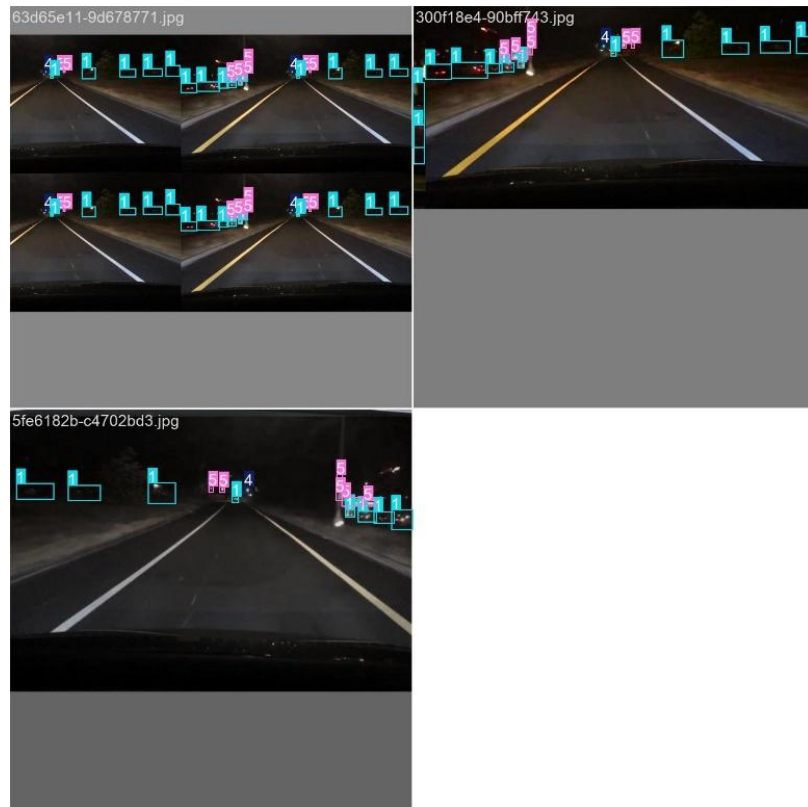


Figure 5.9: ACDC nighttime detection results

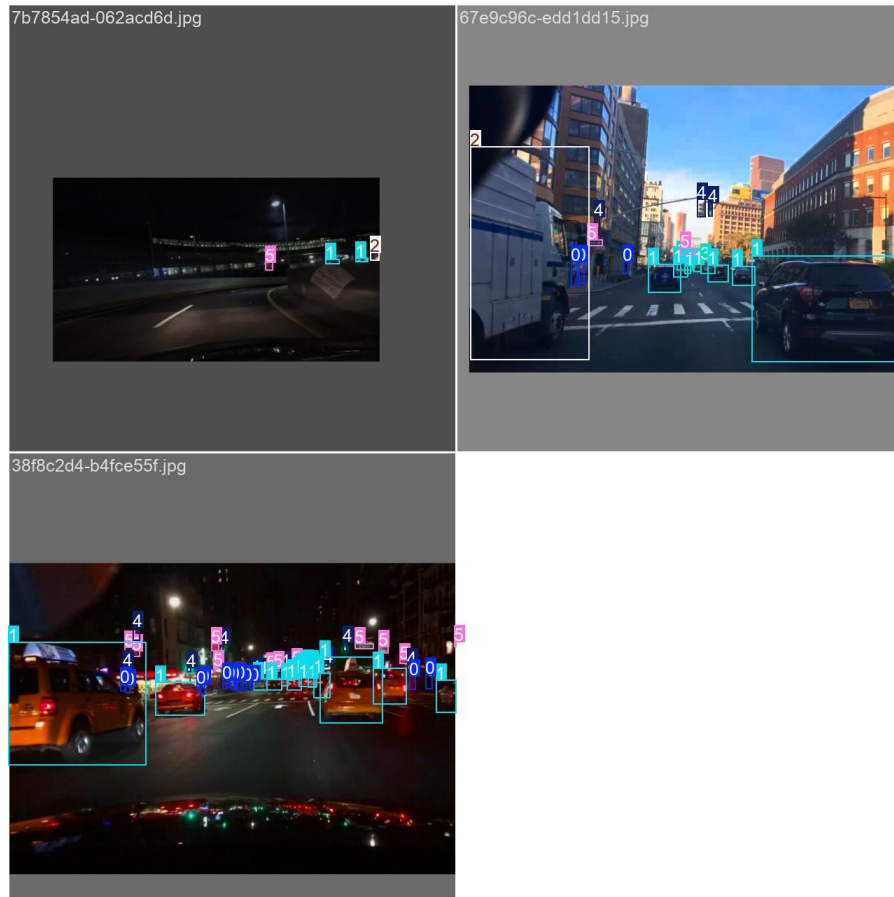


Figure 5.10: ACDC rain and night detection results



Figure 5.11: ACDC foggy and snowy detection results

The qualitative detection outputs confirm the model's capability to locate and classify objects across all four adverse conditions. In nighttime highway scenes, the model successfully detects traffic signs and vehicles at distances where ambient illumination is near-absent. In rain conditions, pedestrian and car detections are maintained despite windshield distortion and road surface reflections. Confidence scores, while lower than those observed in clear-condition BDD100K scenes (as expected), remain above operational thresholds for the majority of detected instances.

CHAPTER 6: DISCUSSION

6.1 Interpretation of Results

The experimental results demonstrate a clear and consistent improvement in detection performance through the two-stage training strategy. The progression from YOLOv8n to YOLOv8m, followed by domain-specific fine-tuning on ACDC, validates the core hypothesis that a unified deep learning approach combining transfer learning with weather-aware augmentation can meaningfully improve object detection across multiple adverse visibility conditions simultaneously, without requiring additional sensors or architectural modifications.

6.1.1 Impact of Model Scale

The upgrade from YOLOv8n to YOLOv8m yielded substantial improvements in Recall (+16.2 pp) and mAP@50 (+16.1 pp). This gain is primarily attributed to the medium model's increased parameter count, which enables the learning of more complex and discriminative feature representations particularly important for objects that appear with reduced contrast, motion blur, or partial occlusion in low-light environments. The F1 score improvement from 0.54 to 0.67 confirms that the medium model achieves a significantly better balance between precision and recall, a critical requirement for safety-sensitive detection tasks where both missed detections and false alarms carry operational consequences.

6.1.2 Effect of ACDC Fine-Tuning

Fine-tuning on ACDC produced qualitatively meaningful improvements in the model's ability to detect objects in genuinely adverse scenes. Validation mAP@50 continued to improve across all 20 fine-tuning epochs, suggesting that the domain adaptation process was not saturated and that extended training would likely yield further performance gains. The relatively short fine-tuning duration was a practical constraint imposed by the dataset size (~3,000 images) and available GPU memory rather than a methodological choice, and represents a recognized opportunity for improvement in Phase 4.

6.1.3 Per-Class Analysis

The Car class consistently achieved the highest detection performance across all configurations, which is expected given its dominant instance count in both datasets. Pedestrian detection, while substantially improved by the model upgrade, remained comparatively lower in recall consistent with findings reported in prior literature

attributing this to the challenge of detecting small, partially occluded human figures against dark or precipitation-obscured backgrounds. The Bus class showed improvement following model scaling and fine-tuning, demonstrating the benefit of greater representational capacity for minority classes.

Traffic Light detection at mAP@50 of 0.666 is particularly notable given that traffic lights are small, frequently occluded objects that present significant difficulty under adverse weather. Reliable detection of traffic control infrastructure under low-visibility conditions has direct safety relevance for autonomous driving deployment.

6.1.4 Confusion Matrix Insights

The normalized confusion matrices reveal that background misclassification is the dominant error mode for pedestrians and buses — a well-documented challenge in adverse-condition detection where foreground-background contrast is fundamentally reduced by fog, rain-induced haze, or nighttime illumination. The persistent Car-Truck inter-class confusion (9–17% of instances) reflects visual similarity at medium and far detection ranges and is expected to decrease with additional fine-tuning data and extended training.

6.1.5 Real-Time Performance

Inference speeds of 2.5–4.7 ms per image were recorded across both model variants, confirming suitability for real-time deployment. At these speeds the system can process well over 200 frames per second — far exceeding standard automotive sensor rates of 30–60 FPS. This is particularly significant because many prior approaches to adverse-condition detection rely on computationally expensive multi-sensor fusion or image enhancement preprocessing that substantially increases inference latency, reducing their viability for real-time embedded deployment.

6.2 Comparison with Literature

The results obtained are contextually consistent with findings reported in the reviewed literature. Studies such as Kumar et al. [2] and Chen et al. [6] identified significant detection performance degradation under adverse weather using standard YOLO variants, which is directly reflected in the YOLOv8n baseline results. The improvement achieved through domain-specific fine-tuning aligns with the approach validated by Huang et al. [1], which similarly employed weather-aware training on ACDC to improve YOLO-family detection robustness. Unlike the approaches of Li et al. [4] and Yang et al. [9], which rely on computationally expensive defogging modules or additional sensing modalities, this project demonstrates meaningful improvement through training

strategy alone reinforcing the practical viability of data-centric methods for low-visibility detection.

6.3 Limitations

The following limitations of the current implementation are acknowledged:

- **Training Duration Constraints:** ACDC fine-tuning was limited to 20 epochs. Extended training is expected to yield further gains, particularly for minority classes such as Bus and Rider.
- **Dataset Scale Imbalance:** ACDC contains only 4,006 images compared to BDD100K's 100,000. This disparity may limit the depth of domain adaptation achievable in a single fine-tuning run.
- **Class Instance Imbalance:** Significant imbalance between Pedestrian and other classes (91,349 vs. 7,210 for Bus in BDD100K) contributes to lower recall for underrepresented classes despite augmentation strategies.
- **Single Modality:** The system operates exclusively on RGB camera input. Integration of complementary sensor modalities (LiDAR, radar) could substantially improve performance under extreme conditions.
- **Snow Condition Evaluation:** Detailed quantitative evaluation under the snow-specific subset of ACDC is planned for the final project phase.

CHAPTER 7: SUMMARY AND CONCLUSION

This project addressed a well-recognized gap in autonomous driving perception: the degradation of object detection model performance under real-world adverse visibility conditions including fog, rain, night, and snow. Through a structured two-stage training pipeline built on the YOLOv8 framework, the project demonstrates that a unified, camera-only deep learning approach can meaningfully improve detection robustness across multiple low-visibility conditions simultaneously, without requiring additional sensors, architectural modifications, or computationally expensive preprocessing modules. The key findings are as follows:

- **Model Scale Matters for Low-Visibility Generalization:** Upgrading from YOLOv8n to YOLOv8m produced a 16.1 pp gain in mAP@50 and a 16.2 pp improvement in Recall, confirming that greater representational capacity is critical for learning discriminative features in visually degraded conditions.
- **Sequential Domain Adaptation is Effective:** Fine-tuning on ACDC following BDD100K training successfully adapted the model to adverse-condition imagery, with validation mAP@50 continuing to improve across all 20 fine-tuning epochs, indicating the process was not saturated.
- **Weather-Aware Augmentation Enhances Robustness:** HSV augmentation, mosaic, erasing, and spatial transformations improved generalization to unseen adverse conditions by exposing the model to simulated variations in illumination, contrast, occlusion, and scale.
- **Real-Time Performance is Preserved:** Inference speeds of 2.5–4.7 ms per image confirm that the improved model remains fully suitable for real-time deployment in safety-critical autonomous driving applications.
- **Background Misclassification Remains the Dominant Failure Mode:** Missed detections across all classes are primarily caused by misclassification as background, particularly for pedestrians and buses, motivating future work on attention-based feature enhancement for low-contrast foreground regions.

The project successfully completes Phases 1 through 3 of the four-phase plan. Phase 4 work remaining includes comprehensive per-condition evaluation across fog, rain, night, and snow subsets of ACDC, FPS benchmarking under inference conditions, and finalization of all comparative tables for the report and presentation.

Future directions include extending ACDC fine-tuning duration, exploring knowledge distillation to maintain accuracy at reduced model sizes, and investigating lightweight image enhancement preprocessing such as histogram equalization or learned denoising as a pre-detection step to further improve performance under extreme low-light and heavy fog conditions.

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